

10-9-25 Research Progress

Signal Processing & ML Infrastructure Development

J.C. Vaught

Undergraduate Assistants: Samuel Cancilla and Nathan Kirk

Department of Mechanical Engineering

Research Summary

Graduate Student

J.C. Vaught – M.S. Student Mechanical Engineering

- Machine learning infrastructure development
- GPU sharding and distributed training optimization
- 3 Pulse Length MAE preliminary results

Undergraduate Research Assistants

Samuel Cancilla, 2nd year

- ST-DBSCAN implementation
- Object tracking algorithms- not included

Nathan Kirk, 4th year

- CFAR algorithm development- not included
- Data collection and processing- not included

Section I:

Radar Signal Processing

ST-DBSCAN Algorithm Overview

Spatio-Temporal Density-Based Spatial Clustering

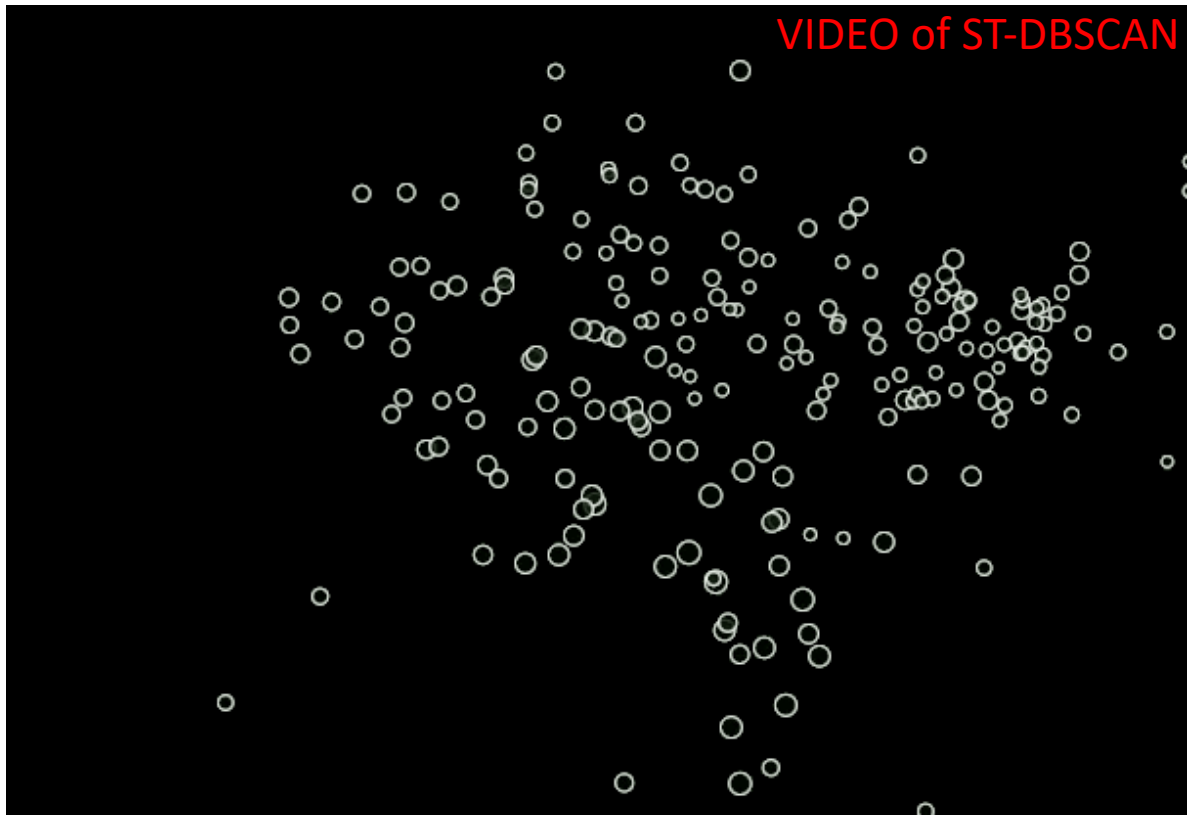
Key Characteristics

- Input: (x, y, t) coordinates
- Parameters: ϵ_1 (spatial), ϵ_2 (temporal)
- Ideal for tracking moving objects
- Handles evolving datasets

Current Challenges

- Computational cost: ~2 hours per 10 frames
- Memory-intensive processing
- Parameter tuning complexity

VIDEO of ST-DBSCAN



ST-DBSCAN Results

Detection Performance

Key Findings

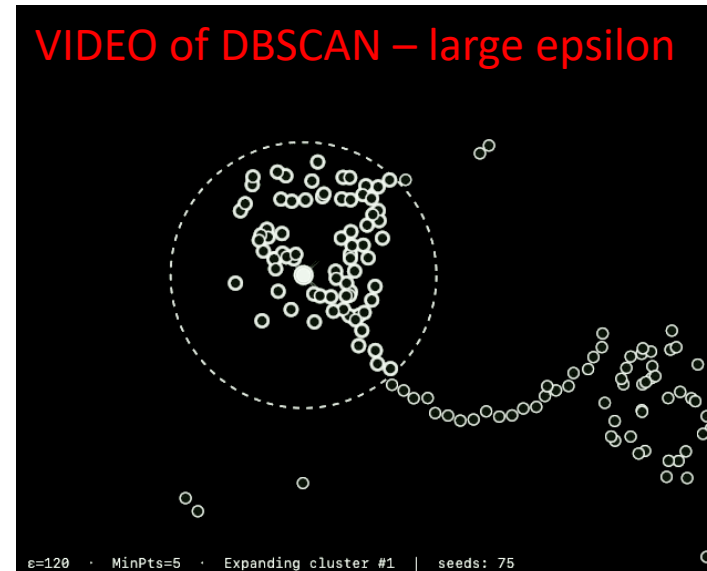
- Tracking IDs persist across frames
- Validation confirms accuracy

Processing Time: ~2 hrs per 10 frames (needs optimization)

VIDEO of DBSCAN – small epsilon



VIDEO of DBSCAN – large epsilon



Section II:

Machine Learning

Infrastructure

Data Pipeline

Source & Sample Generation

- PNG sequences with alternating gain levels
- Locate triplets: gain-65 → gain-60 → gain-70
- 1024×1024 resolution maintained throughout

Preprocessing Options

- **Tile mode:** Crop into four quadrants (4× samples, tighter spatial focus)
- **Quadrant targets:** 8 outputs per sample (2 time steps × 4 quadrants)
- RGBA handling via solid background compositing

Augmentations

- Shared transformations (flips, rotations, jitter, blur, noise) applied to all frames

Model Architecture (Vision-Transformer MAE)

Encoder: ViT-style Stack

- Patch embed: 16×16 stride $\rightarrow$ 4096 patch tokens (+ class token)
- Transformer encoder blocks (multi-head attention + MLP)
- Learnable positional embeddings
- Optional random masking of patch tokens (CLI-toggled)

Decoder: Lighter ViT

- Projects encoder output, appends learned mask tokens via `ids_restore`
- Transformer decoder blocks + linear head predict per-patch pixels
- Unpatchify reassembles predictions into full images (2 or 8 targets)

Training Objective

$$\text{Loss} = \text{MSE} + \lambda \cdot \text{L1}$$

Loss Components

- **MSE** maintains global structure; **L1** encourages crisp edges ($\lambda = 0.1$ default, try 0.2)
- For masked training: loss evaluated only on visible patches

Optimization Setup

- **Optimizer:** AdamW with cosine decay LR
- **Framework:** Hugging Face Accelerate + DeepSpeed ZeRO-2, FP16 precision
- **Hardware:** Single RTX 6000 Ada with gradient accumulation

Next Steps & Extensions

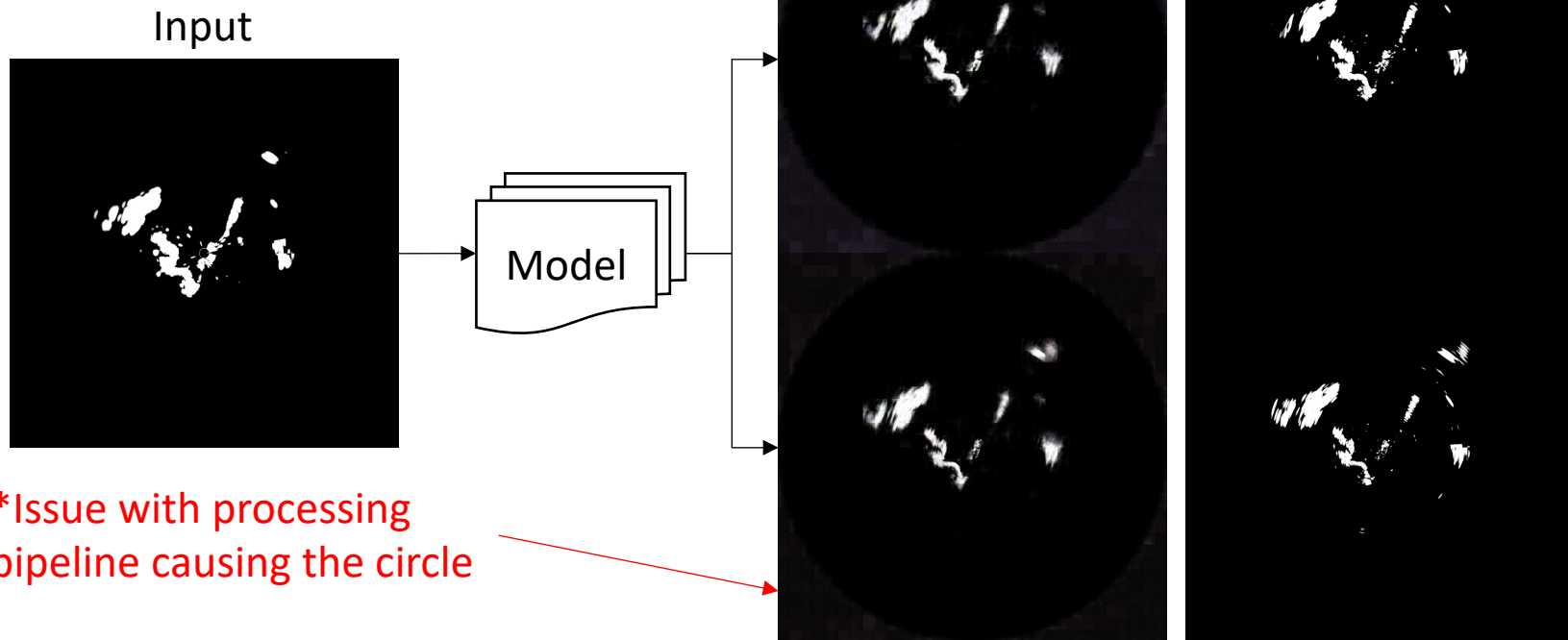
Potential Improvements

- **Mask ratio tuning:** Systematic exploration of optimal masking percentage
- **Perceptual losses:** Incorporate VGG or LPIPS for better perceptual quality
- **Conv refinement head:** Add convolutional layers for final detail enhancement
- **Temporal attention:** Explicit modeling of temporal dependencies
- **Multi-scale architecture:** Process patches at different resolutions
- **Extended prediction:** Predict more than two future frames

Framework adapts He et al.'s MAE to temporal frame prediction

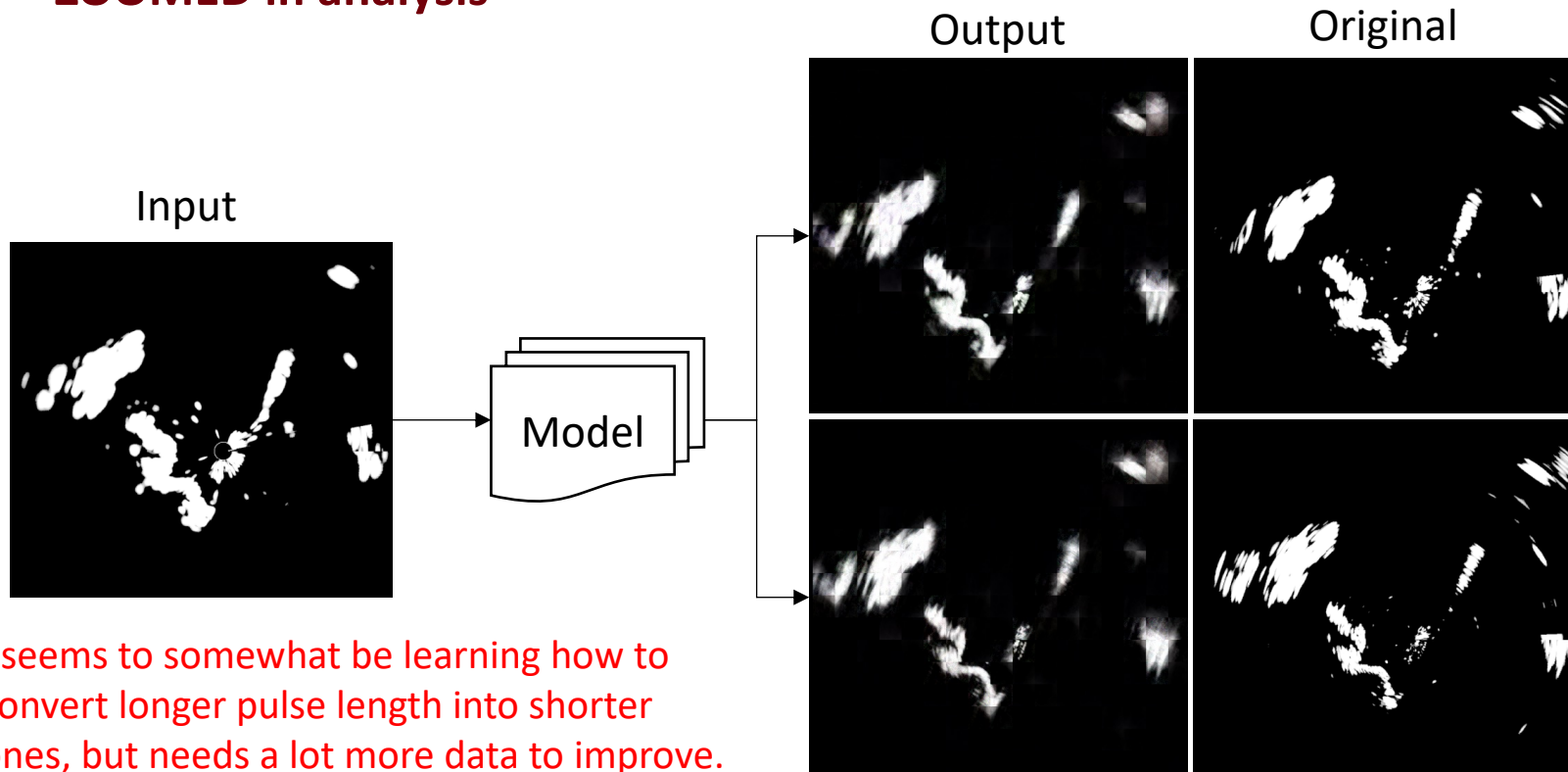
Temporal-MAE Architecture

MAE for Radar Pulse Length



Temporal-MAE Architecture

ZOOMED in analysis

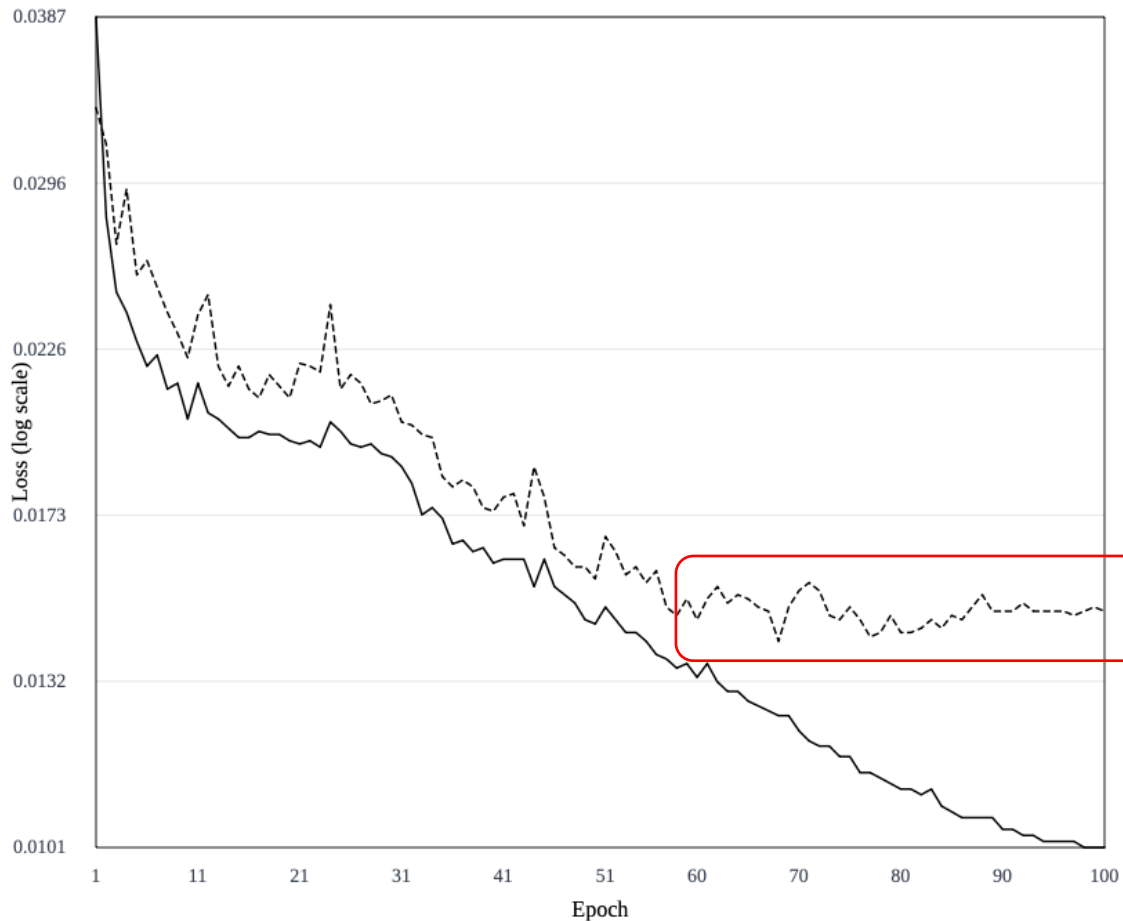


Temporal-MAE Architecture

MAE for Radar Pulse Length Prediction

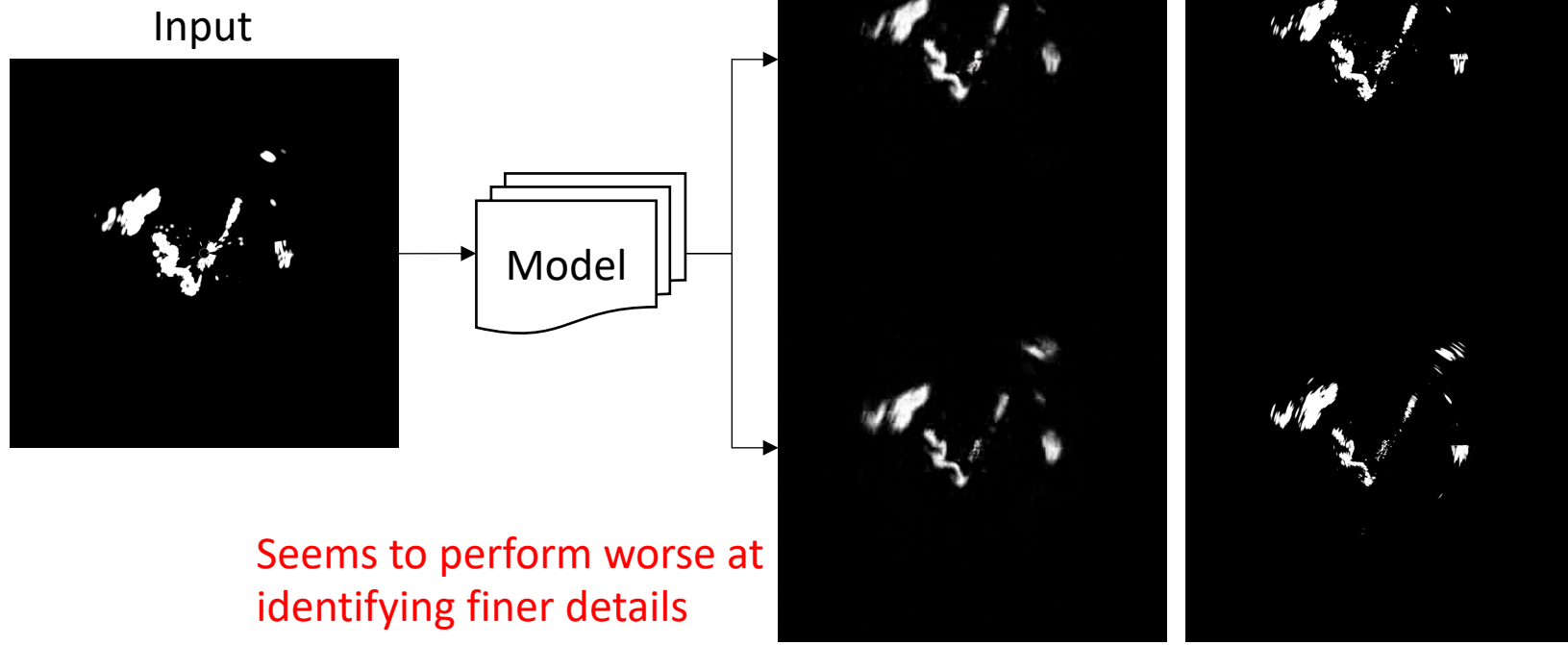
Having issue with overfitting.

Trying an Augmented version now(colors, noise, rotations, etc)



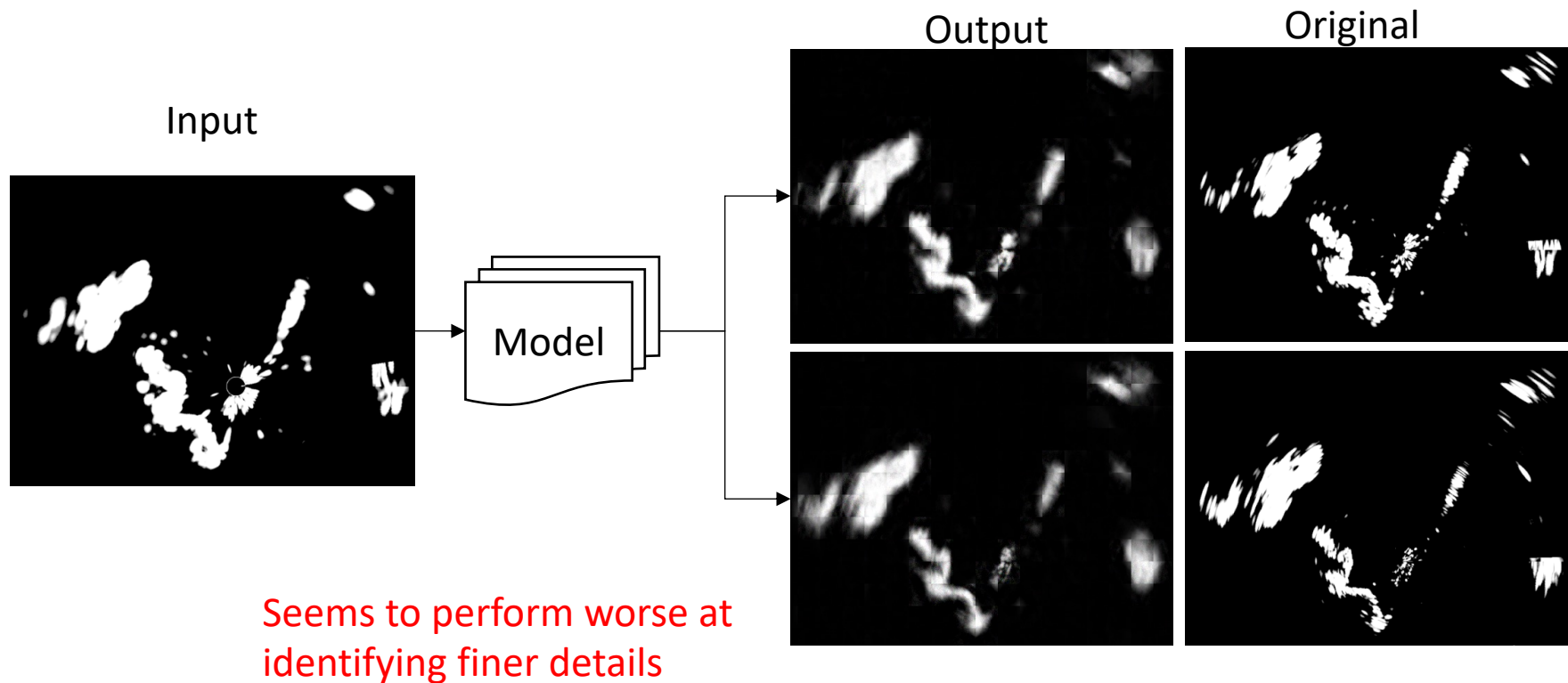
Temporal-MAE Architecture

**MAE for Radar Pulse Length
augmented**



Temporal-MAE Architecture

Zoomed analysis

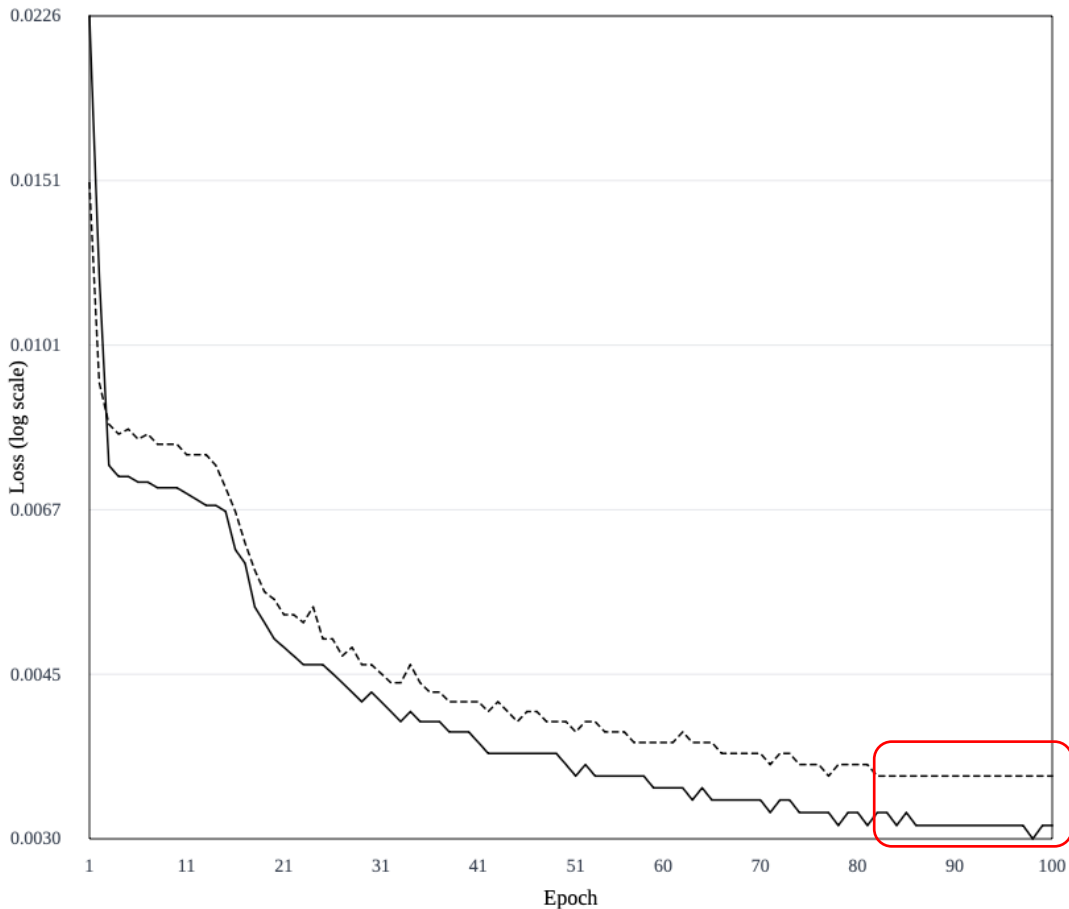


Temporal-MAE Architecture

MAE for Radar Pulse Length Prediction

Worked better* than the prior model.

Trying smaller patch size, different loss functions, and splitting image into quadrants(not complete yet)



Questions?

J.C. Vaught

Department of Mechanical Engineering
University of South Carolina